CHAPTER 11

ADVANCES IN COMPUTING

CHAPTER ESSENCE AT A GLANCE:
Explore emerging paradigms in computing: distributed systems, cloud computing tiers (IaaS, PaaS, SaaS), grid clusters, artificial intelligence, and IoT.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Distributed Computing Architectures:** Distributed paradigms combine independent processing units across networks, distinguishing between loosely federated heterogeneous grids solving immense computational problems and tightly unified homogeneous clusters acting as single supercomputers.

• **Cloud Computing Service Stack:** Cloud computing delivers on-demand virtualized computing resources categorized into infrastructure providing raw computing power, platforms delivering development frameworks, and software offering ready-to-use end-user applications.

• **Big Data Dimensions:** Modern organizational data environments are characterized by massive storage scale, rapid streaming velocity, structural diversity across formats, and the paramount necessity for data veracity and quality.

• **Artificial Intelligence and Machine Cognition:** Artificial intelligence develops computational algorithms capable of performing tasks typically requiring human intelligence, including computer vision, natural language understanding, and automated decision making.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{SaaS (Applications)} \longrightarrow \text{PaaS (Platforms)} \longrightarrow \text{IaaS (Hardware/VMs)}$$

Cloud computing tiered service delivery architecture

$$\text{Big Data} = \text{Volume} + \text{Velocity} + \text{Variety} + \text{Veracity}$$

The four fundamental governing V dimensions of Big Data

$$\text{Grid Computing (Heterogeneous)} \quad \longleftrightarrow \quad \text{Cluster Computing (Homogeneous)}$$

High-performance distributed computing architecture comparison

$$\text{Latency Reduction} = \frac{\text{Cloud Roundtrip Time} - \text{Edge Response Time}}{\text{Cloud Roundtrip Time}} \times 100\%$$

Edge computing latency improvement metric over remote cloud

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** SaaS provides complete ready-to-use software applications over the web, whereas IaaS delivers virtualized raw hardware infrastructure.

• **Alert:** Public clouds offer shared multi-tenant resources, private clouds dedicate infrastructure to single organizations, and hybrid clouds blend both.

• **Alert:** Grid computing harnesses heterogeneous loosely coupled machines, unlike specialized tightly coupled supercomputer clusters.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Differentiate between Distributed Computing, Grid Computing, and Cluster Computing. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Distributed Computing):** Autonomous computing systems on a network communicating by message-passing to solve large problems. **[1 Mark]**

• **Step 2 (Grid Computing):** Geographically dispersed, heterogeneous computing resources pooled together over Internet to solve complex scientific calculations. **[1 Mark]**

• **Step 3 (Cluster Computing):** Dedicated homogeneous computers connected by high-speed LAN acting as a single high-performance computer. **[1 Mark]**

• **Key Exam Result:** Distributed (networked), Grid (heterogeneous internet pool), Cluster (homogeneous LAN)

$$\text{Grid Computing (Heterogeneous)} \quad \longleftrightarrow \quad \text{Cluster Computing (Homogeneous)}$$

High-performance distributed computing architecture comparison

2. What is Cloud Computing? Explain the three cloud service models: IaaS, PaaS, and SaaS. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Cloud Computing):** On-demand delivery of computing resources (servers, storage, databases, networking, software) over the Internet with pay-as-you-go pricing. **[1 Mark]**

• **Step 2 (Service Models):** 1. IaaS (Infrastructure as a Service): Provides raw virtual machines, storage, and networking (e.g. AWS EC2, Google Compute Engine). 2. PaaS (Platform as a Service): Provides environment/runtime to develop and deploy apps without managing servers (e.g. Google App Engine, Heroku). 3. SaaS (Software as a Service): Delivers ready-to-use software applications over web (e.g. Google Workspace, Gmail, Microsoft 365). **[3 Marks]**

• **Key Exam Result:** IaaS (hardware) --> PaaS (platform) --> SaaS (software)

$$\text{SaaS (Applications)} \longrightarrow \text{PaaS (Platforms)} \longrightarrow \text{IaaS (Hardware/VMs)}$$

Cloud computing tiered service delivery architecture

3. Explain the four cloud deployment models: Public, Private, Hybrid, and Community Cloud. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Deployment Models):** 1. Public Cloud: Owned by third-party provider, shared by multiple organizations over internet (AWS, Azure). 2. Private Cloud: Dedicated exclusively to a single business/organization, maximum security. 3. Hybrid Cloud: Combines public and private clouds, allowing data sharing between them. 4. Community Cloud: Shared by organizations with common interests (e.g. banks, universities). **[3 Marks]**

• **Key Exam Result:** Public, Private, Hybrid, Community clouds

$$\text{SaaS (Applications)} \longrightarrow \text{PaaS (Platforms)} \longrightarrow \text{IaaS (Hardware/VMs)}$$

Cloud computing tiered service delivery architecture

4. What is Artificial Intelligence (AI)? Name and explain its core sub-fields (Machine Learning, NLP). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (AI Definition):** Branch of computer science that builds smart machines capable of performing tasks that typically require human intelligence (reasoning, learning, problem solving). **[1 Mark]**

• **Step 2 (Subfields):** Machine Learning (algorithms learning from data), Deep Learning (neural networks), Natural Language Processing (understanding text/speech), Computer Vision, Robotics. **[2 Marks]**

• **Key Exam Result:** AI: Machine Learning, Deep Learning, NLP, Computer Vision

$$\text{SaaS (Applications)} \longrightarrow \text{PaaS (Platforms)} \longrightarrow \text{IaaS (Hardware/VMs)}$$

Cloud computing tiered service delivery architecture

5. Explain the Four V's of Big Data: Volume, Velocity, Variety, and Veracity. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Four V's of Big Data):** 1. Volume: Massive scale and quantity of data generated (terabytes to petabytes). 2. Velocity: High speed of data generation and real-time streaming (IoT, social media). 3. Variety: Different formats: structured (tables), semi-structured (JSON/XML), unstructured (videos, audio). 4. Veracity: Trustworthiness, accuracy, and quality of data. **[3 Marks]**

• **Key Exam Result:** Four V's: Volume, Velocity, Variety, Veracity

$$\text{Big Data} = \text{Volume} + \text{Velocity} + \text{Variety} + \text{Veracity}$$

The four fundamental governing V dimensions of Big Data

6. What is the Internet of Things (IoT)? Explain its architecture and real-world applications. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (IoT Definition):** Network of physical devices ('things') embedded with sensors, software, and connectivity that collect and exchange data over the internet without human intervention. **[1.5 Marks]**

• **Step 2 (Applications):** Smart homes (automated lighting, thermostats), smart agriculture (soil moisture sensors), smart healthcare (wearable ECG monitors), smart cities (traffic management). **[1.5 Marks]**

• **Key Exam Result:** IoT: interconnected smart devices exchanging sensor data

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Cloud computing tiered service delivery architecture

7. Discuss the advantages and challenges of adopting cloud computing for businesses. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Cloud Advantages):** 1. Cost savings (no expensive hardware purchases). 2. Global scalability and elasticity on demand. **[1 Mark]**

• **Step 2 (Security Concerns):** 1. Data privacy risks (data stored on third-party servers). 2. Dependency on reliable high-speed internet connection. **[1 Mark]**

• **Key Exam Result:** Advantages: Scalability & cost saving | Concerns: Privacy & internet dependency

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Cloud computing tiered service delivery architecture